Homework 1

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Problem 1

(a) The map $B \circ A = C : \mathbb{Z} \to \mathbb{N}$ is defined by

$$C(n) = B(n+1) = (n+1)^2 - n - 1 = n(n+1).$$

The image is $C(\mathbb{Z}) = \left\{n^2 + n \in \mathbb{N} \mid n \in \mathbb{Z}\right\}$.

(b) The map A is a bijection. Let $n \in \mathbb{Z}$. Since $\mathbb{Z}$ is closed under addition/subtraction, we have $n - 1 \in \mathbb{Z}$, and thus $A(n-1) = n$. Therefore, A is surjective. Now suppose that $A(n) = A(m)$ for some $n, m \in \mathbb{Z}$. By definition of A,

$$A(n) = A(m) \implies n + 1 = m + 1 \implies n + 1 - 1 = m + 1 - 1 \implies n = m.$$

Therefore, A is injective, and thus bijective.

The map B is neither. Note that

$$B(2) = 4 - 2 = 2 = 1 + 1 = (-1)^2 - (-1) = B(-1).$$

Since $2 \neq -1$, B is not injective. Further, suppose $3 = n(n+1)$. We have

$$n^2 + n - 3 = 0 \implies n = \frac{-1 \pm \sqrt{1 + 12}}{2} = \frac{-1 \pm \sqrt{13}}{2} \notin \mathbb{Z}.$$

Therefore, there exists at least one element $3 \in \mathbb{N}$ such that $3 \notin B(\mathbb{Z})$, and thus B is not surjective.

Finally, note that since $A(\mathbb{Z}) = \mathbb{Z}$, we have that $B(\mathbb{Z}) = B(A(\mathbb{Z})) = C(\mathbb{Z})$, so C is not surjective. Additionaly,

$$C(1) = 2 = (-1)^2 + (-1) = C(-1).$$

Therefore, C is not injective either.